Notes version: 14 Nov 22:41 IST (if this time is old, reload the page)

GEOGRAPHY · Transport and communication

UNDERSTAND IT FIRST

What is it? Transport moves **people and goods** from where they are to where they are needed; **communication** moves **messages and information**. Together they are the **'lifelines' of the economy**: a farmer's chillies reach a Guntur market by road, then a port by rail, then another country by ship. Each mode has a strength: **roads** reach every village (door-to-door, short distance), **railways** carry **bulk goods and crowds** over long distances cheaply, **waterways** are the **cheapest** for heavy cargo, **pipelines** carry oil and gas without stopping, and **air** is the **fastest** but costliest. A good network lowers **logistics cost** (India's is high, around **13–14% of GDP** by older estimates; the 2023 NCAER study put it near **8–9%**). Learn it as **roads → railways → waterways and ports → air and pipelines → communication → Andhra Pradesh**.

Five modes of transport: strengths

1. Roads

- **Network:** India has the **second-largest road network** in the world (about **63 lakh km**, after the USA). **National Highways (NH)** are only about **2%** of road length but carry about **40%** of traffic. NH length grew from about **91,287 km (2014)** to about **1,46,572 km (December 2025)**; about **5,364 km** of **high-speed (access-controlled) corridors** are open. NHs are built by the **Union** through **NHAI (1988 Act, set up 1995)**, NHIDCL and the **Ministry of Road Transport and Highways**; **State Highways** by States (PWD); **district and rural roads** by local bodies and States.
- **Big projects: Golden Quadrilateral** (about **5,846 km**; links **Delhi–Mumbai–Chennai–Kolkata**; **NHDP 1998–99**, under PM Vajpayee); **North–South corridor** (**Srinagar–Kanyakumari**) and **East–West corridor** (**Silchar–Porbandar**) meet at **Jhansi**; **Bharatmala Pariyojana (2017)**: economic corridors, border and coastal roads, expressways; **PM Gram Sadak Yojana (PMGSY, 25 December 2000)** for all-weather roads to villages; **Setu Bharatam** (rail over-bridges). Longest NH: **NH-44** (**Srinagar–Kanyakumari**, about **4,100 km**). Expressways: **Delhi–Mumbai Expressway**, **Mumbai–Pune** (first, **2002**), **Purvanchal**, **Samruddhi**.
- **Odd-even numbering of NHs (2010 system): north–south** NHs carry **even** numbers that rise from **east to west**; **east–west** NHs carry **odd** numbers that rise from **north to south**. So NH-44 runs north–south and NH-16 (the **east-coast highway** through **Srikakulam–Visakhapatnam–Vijayawada–Nellore** to Chennai) runs along the east. **Border Roads Organisation (BRO, 1960)** builds roads in border areas: **Atal Tunnel (Rohtang, 2020)**, the **Umling La** road (among the world's highest motorable roads).

2. Railways

- **History:** first passenger train **Bombay (Bori Bunder)–Thane, 16 April 1853** (34 km; Lord **Dalhousie**); in the south, **Royapuram–Wallajah Road (1856)**. **Nationalised in stages**; **Railway Board 1905**. The **Railway Budget** was **separate from 1924** (**Acworth Committee 1920–21**) and **merged with the Union Budget in 2017**.

- **Size:** about **69,439 route km (March 2025)**; **fourth-largest** network (after the USA, China, Russia) and among the **largest employers**. Electrification about **99%+** of broad gauge (**99.6% in 2025–26**); about **741 crore** passengers carried in 2025–26. Gauges: **broad (1.676 m)**, **metre (1 m)**, **narrow (0.762/0.610 m)**; **Project Unigauge (1992)** converts to broad gauge. Hill railways **Darjeeling (1999), Nilgiri (2005), Kalka–Shimla (2008)** are **UNESCO World Heritage** (together the 'Mountain Railways of India').
- **Zones: 18 zones** now: the **South Coast Railway (SCoR)**, headquarters **Visakhapatnam**, began working on **1 June 2026** (Waltair division reorganised; Indian Railways' logo now has **18 stars**). Others to remember: **Northern (New Delhi)**, **South Central (Secunderabad)**, **East Coast (Bhubaneswar)**, **Southern (Chennai)**, **Konkan Railway** (a separate corporation, **1998**, 741 km along the west coast).
- **Modern projects: Dedicated Freight Corridors**: **Eastern DFC (Ludhiana–Sonnagar, ~1,337 km)** and **Western DFC (Dadri–JNPT, ~1,506 km)**; about **2,741 km** commissioned by October 2025. **Vande Bharat** semi-high-speed trains (**2019**, made at **ICF Chennai**); **Mumbai–Ahmedabad High-Speed Rail ('bullet train')** with **Japan** (~508 km); **Kavach** (home-made anti-collision system); **Amrit Bharat Station Scheme**; the **Chenab bridge** (world's highest rail arch bridge) and **Anji Khad** bridge on the **USBRL** line to Kashmir (2025); **Pamban** vertical-lift sea bridge (2025). Factories: **CLW Chittaranjan** (electric locos), **BLW Varanasi**, **ICF Chennai**, **RCF Kapurthala**, **Rail Wheel Factory Yelahanka**.

3. Waterways, ports, air and pipelines

Mode	Key facts
Inland waterways	**Cheapest and greenest** for bulk cargo. **National Waterways Act 2016** declared **111 NWs**; about **32** are working (about **5,155 km**, Nov 2025). Cargo rose from **18 MMT (2013–14)** to about **146 MMT (2024–25)**. **IWAI (1986, Noida)** manages them. **NW-1: Ganga–Bhagirathi–Hooghly, Haldia–Prayagraj (~1,620 km), longest**; **NW-2: Brahmaputra (Dhubri–Sadiya)**; **NW-3: West Coast Canal (Kerala)**; **NW-4: Krishna–Godavari** with the **Buckingham Canal** (AP–Tamil Nadu); **NW-5**: East Coast Canal with Brahmani–Mahanadi. **Jal Marg Vikas** project (World Bank) on NW-1.
Ports and shipping	About **95% of trade by volume** (about 70% by value) goes by sea. **12 major ports** (Union, **Major Port Authorities Act 2021**) plus about **200 non-major** ports (States). West coast: **Deendayal (Kandla**, tidal, largest by cargo for years), **Mumbai**, **JNPA (Nhava Sheva**, biggest container port), **Mormugao**, **New Mangalore**, **Kochi**. East coast: **V.O. Chidambaranar (Tuticorin)**, **Chennai**, **Kamarajar (Ennore**, corporate port), **Visakhapatnam** (deepest natural harbour on the east coast, protected by **Dolphin's Nose**), **Paradip**, **Syama Prasad Mookerjee (Kolkata**, riverine, with **Haldia** dock). **Vadhavan** (Maharashtra) is being built as a mega deep-water port (13th major port). **Sagarmala (2015)** for port-led growth; **Maritime India Vision 2030**, **Amrit Kaal Vision 2047**. Private ports: **Mundra** (largest private), **Krishnapatnam**, **Gangavaram**.
Air transport	India is the **third-largest domestic aviation market**. About **412 million** airport passengers in **FY25**. **Airports Authority of India (AAI, 1995)** runs most airports; **DGCA** regulates safety; **AERA** sets tariffs; **BCAS** security. **Air India** nationalised **1953**, sold back to **Tata (January 2022)**. **UDAN** (*Ude Desh ka Aam Nagrik*, **2016**) gives regional connectivity with viability-gap funding; the airport count roughly **doubled since 2014** (to about **160**). **Pawan Hans** runs helicopters. **Greenfield** airports: Navi Mumbai, Noida (Jewar), **Bhogapuram (Vizag)**.

Pipelines	Carry **crude, products and gas** cheaply, safely and without stopping; costly to build. First crude line **Naharkatiya–Nunmati–Barauni (Assam–Bihar, 1960s)**; **Salaya–Mathura** (crude to Mathura refinery); **HBJ gas pipeline (Hazira–Bijaipur–Jagdishpur, GAIL, 1980s)**; **KG basin East–West pipeline (Kakinada–Bharuch)**; **Pradhan Mantri Urja Ganga** (Jagdishpur–Haldia–Bokaro–Dhamra) for the east; **'One Nation One Gas Grid'**. Cross-border: **India–Bangladesh Friendship Pipeline (2023)**, **Motihari–Amlekhgunj** (Nepal, 2019, first cross-border product line in South Asia).

4. Communication and Andhra Pradesh

Communication in India: milestones

- **Personal communication: India Post** (unified **1854**, Dalhousie; the **world's largest postal network**, about **1.6 lakh post offices**, most in villages); **PIN code** (6 digits) introduced **15 August 1972** (first digit = region; **9** = Army Post Office); **Speed Post 1986**; **India Post Payments Bank (2018)**. **Telegraph** service closed **15 July 2013**. **Telephone**: about **120 crore** subscribers, mostly mobile; **teledensity** ~85%; **5G launched 1 October 2022**; **BharatNet** takes optical fibre to gram panchayats; **Telecommunications Act 2023** replaced the **Indian Telegraph Act 1885**; regulator **TRAI (1997)**; dispute tribunal **TDSAT**.
- **Mass communication:** radio broadcasting began **1927** (Bombay); **Indian State Broadcasting Service 1930 → All India Radio 1936 → 'Akashvani' 1957**; **Doordarshan**: first TV **15 September 1959 (Delhi)**, colour TV and national broadcast at the **1982 Asian Games**; **Prasar Bharati (1997)** runs AIR and DD. Satellites **INSAT/GSAT** made nationwide TV and phones possible. **Press**: **Registrar of Newspapers for India (RNI)**, now under the **Press and Registration of Periodicals Act 2023**; **Press Council of India (1966/1978)**. Films certified by **CBFC**.
- **Andhra Pradesh: NH-16** runs along the whole coast (Ichchapuram to Tada), with links like **NH-44** (Anantapur–Kurnool) and **NH-40** (Kurnool–Kadapa–Chittoor); **Bengaluru–Chennai** and **Raipur–Visakhapatnam** expressways cross the State. Rail: the new **South Coast Railway (Visakhapatnam)**; junctions **Vijayawada** (among the busiest), **Guntakal**, **Renigunta**, **Gudur**; the **Kottavalasa–Kirandul (KK) line** carries **Bailadila iron ore** to Vizag through the **Araku** valley and **Borra** tunnels. Ports: **Visakhapatnam** (major), **Gangavaram**, **Krishnapatnam**, **Kakinada** (deep-water and anchorage), and new **Ramayapatnam**, **Machilipatnam**, **Mulapeta** ports. Airports: **Visakhapatnam**, **Vijayawada (Gannavaram)**, **Tirupati (Renigunta)**, **Rajahmundry**, **Kadapa**, **Kurnool (Orvakal)**, **Puttaparthi**; the new **Alluri Sitarama Raju International Airport, Bhogapuram** (Vizianagaram district) opened in **2026**. **NW-4** (Krishna–Godavari, Buckingham Canal) is in AP. Vijayawada and Vizag **metro rail** projects are planned.

OTHER ANGLES ASKED IN EXAMS

Other angles asked: (1) **PM Gati Shakti (October 2021)**: a GIS master plan joining **16 ministries** for infrastructure planning; **National Logistics Policy (17 September 2022)**; **LEADS** index ranks States on logistics; **Logistics Performance Index (World Bank)**: India **38th (2023)**. (2) **Multimodal logistics parks** and **Inland Container Depots**; **FASTag (mandatory 2021)**. (3) The **Golden Quadrilateral** joins the four metros; the **NS–EW corridors** cross at **Jhansi**. (4) **Kandla** was built after Partition to replace **Karachi**; **Kolkata** is a **riverine** port that needs constant **dredging**; **Kochi** is a natural harbour on a lagoon; **Mormugao** exports

iron ore. (5) **Sea routes**: India's trade passes the **Strait of Malacca**, **Bab-el-Mandeb**, **Hormuz**; the **Suez** route links Europe. (6) **Ro-Ro** ferries, **seaplanes**, **ropeways** (**Parvatmala**, 2022). (7) **Vande Bharat sleeper** and **Namo Bharat (RRTS, Delhi–Meerut)** are new categories.

EXAM TRAP

Roads: 2nd largest (~63 lakh km); NH ~2% of length, ~40% traffic; NH 91,287 km (2014) → 1,46,572 km (Dec 2025); NHAI 1988/1995; GQ 5,846 km (NHDP 1998–99); NS (Srinagar–Kanyakumari) + EW (Silchar–Porbandar) cross at Jhansi; Bharatmala 2017; PMGSY 25 Dec 2000; NH-44 longest; even = N–S, odd = E–W; BRO 1960; Atal Tunnel 2020; Rail: 16 Apr 1853 Bombay–Thane; 1924 separate budget (Acworth), merged 2017; ~69,439 route km; 4th largest; ~99.6% electrified; gauges 1.676/1/0.762 m; Unigauge 1992; UNESCO hill railways; 18 zones (SCoR, Vizag, 1 June 2026); Konkan 1998; EDFC Ludhiana–Sonnagar, WDFC Dadri–JNPT; Vande Bharat 2019; MAHSR with Japan; Kavach; Chenab bridge; KK line; Water: NW Act 2016 (111); ~32 working; NW-1 Haldia–Prayagraj 1,620 km; NW-2 Brahmaputra; NW-3 Kerala; NW-4 Krishna–Godavari; IWAI 1986; cargo 146 MMT; Ports: ~95% trade by volume; 12 major (2021 Act); Kandla tidal; JNPA containers; Vizag deepest (Dolphin's Nose); Kolkata riverine; Vadhavan 13th; Sagarmala 2015; Mundra largest private; Air: 3rd largest domestic; AAI 1995; DGCA; UDAN 2016; Air India → Tata 2022; ~160 airports; Bhogapuram 2026; Pipelines: Naharkatiya–Barauni; HBJ (GAIL); Urja Ganga; India–Bangladesh 2023; Motihari–Amlekhgunj 2019; Post 1854; PIN 1972; telegraph ended 15 July 2013; 5G 1 Oct 2022; Telecom Act 2023; TRAI 1997; AIR 1936, Akashvani 1957; DD 1959; Prasar Bharati 1997; Gati Shakti 2021; NLP 2022.

Test yourself: 8 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:
1. National Highways form only about 2% of India's road length but carry about 40% of road traffic.
2. The North–South and East–West corridors intersect at Jhansi.
3. Under the present numbering system, north–south National Highways carry odd numbers.
4. The Golden Quadrilateral links Delhi, Mumbai, Chennai and Kolkata.
Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. Statement 3 is wrong: north–south NHs carry EVEN numbers (e.g. NH-44); east–west NHs carry odd numbers. The rest are correct.

PRACTICE

Match List-I (major port) with List-II (feature):

List-I (major port)	List-II (feature)
A. Deendayal (Kandla)	1. Riverine port that needs constant dredging
B. Jawaharlal Nehru Port	2. Tidal port built after Partition to replace Karachi
C. Visakhapatnam	3. Largest container port
D. Syama Prasad Mookerjee (Kolkata)	4. Deepest natural harbour on the east coast, sheltered by Dolphin's Nose

(1) A → 2 B → 3 C → 1 D → 4

(2) A → 3 B → 2 C → 4 D → 1

(3) A → 2 B → 3 C → 4 D → 1

(4) A → 2 B → 4 C → 3 D → 1

Answer: (3) Kandla: tidal, post-Partition; JNPA: largest container port; Vizag: deep natural harbour, Dolphin's Nose; Kolkata: riverine, dredging.

PRACTICE

Which one of the following statements about Indian Railways is NOT correct?

A. The first passenger train ran between Bombay and Thane in 1853.

B. The Railway Budget was separated from the General Budget in 1924 on the Acworth Committee's advice.

C. The Kalka–Shimla, Nilgiri and Darjeeling railways are UNESCO World Heritage properties.

D. The headquarters of the South Coast Railway zone is at Vijayawada.

Answer: D. South Coast Railway's headquarters is Visakhapatnam; the zone began working on 1 June 2026 as the 18th zone.

PRACTICE

Assertion (A): Inland water transport is the cheapest mode for moving bulk cargo over long distances.
Reason (R): Waterways need no costly track or road building and a barge carries large loads using little

fuel per tonne-km.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Both true, and R explains A: low building cost and high fuel efficiency make waterways cheapest.

PRACTICE

A railway line in Andhra Pradesh climbs through the Araku valley and Borra tunnels and mainly carries iron ore from Bailadila (Chhattisgarh) to Visakhapatnam port. Which line is it?

A. Guntakal–Renigunta line

B. Kottavalasa–Kirandul line

C. Vijayawada–Gudur line

D. Nandyal–Yerraguntla line

Answer: B. The Kottavalasa–Kirandul (KK) line carries Bailadila ore to Vizag through the Araku valley.

PRACTICE

Consider the following statements:

I. National Waterway-1 runs on the Ganga–Bhagirathi–Hooghly system between Haldia and Prayagraj.

II. National Waterway-4 includes the Krishna and Godavari rivers along with the Buckingham Canal.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Answer: C. Both are correct.

PRACTICE

Consider the following pairs (pipeline/project – what it carries or links):

1. HBJ pipeline – natural gas from Hazira

2. Naharkatiya–Barauni – crude oil from Assam

3. Motihari–Amlekhgunj – petroleum products to Nepal

4. Pradhan Mantri Urja Ganga – crude oil from Mumbai High to Kolkata

How many of the above pairs are correctly matched?

A. Only one pair

B. Only two pairs

C. Only three pairs

D. All four pairs

Answer: C. Pair 4 is wrong: Urja Ganga is a GAS pipeline (Jagdishpur–Haldia–Bokaro–Dhamra) for eastern India. Pairs 1–3 are correct.

PRACTICE

Arrange the following in chronological order:

1. PIN code system introduced

2. Telegraph service closed

3. Doordarshan's first telecast in Delhi

4. 5G services launched

Select the correct order:

A. 1, 3, 2, 4

B. 3, 1, 4, 2

C. 3, 2, 1, 4

D. 3, 1, 2, 4

Answer: D. Doordarshan 1959 → PIN 1972 → telegraph closed 2013 → 5G 2022.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the modes with one strength each; roads (network, NH facts, GQ, corridors, Bharatmala, PMGSY, numbering). 2. Write railways (1853, budgets, size, gauges, zones incl. SCoR, DFCs, new projects,

factories). 3. Write waterways (NW 1–5), major ports east and west, air bodies and UDAN, pipelines. 4. Write communication milestones and AP's roads, rail, ports and airports.

ECONOMY · Indian agriculture: cropping pattern, production and productivity

UNDERSTAND IT FIRST

What is it? Three words that exams mix up. **Cropping pattern** = **which crops are grown, and in what share of the land**, in a place at a time (for example, how much of the cropped area is under rice, wheat, pulses, cotton). **Production** = the **total quantity** harvested (million tonnes). **Productivity (yield)** = **output per hectare** (tonnes or kg per hectare). Production can rise because **area** rises or because **yield** rises; India's growth since the **Green Revolution** came mainly from **yield**. Agriculture still employs about **45% of workers** but gives only about **16–18% of GVA**, so raising **productivity** is the key to raising farm incomes. Learn it as **land use → cropping pattern and its causes → production record → productivity and why it is low → policies → Andhra Pradesh**.

1. Land use and cropping pattern

From land to cropped area

- **Land use (nine-fold classification):** of India's reporting area (about **30.5 crore ha**), the **net sown area** is about **14 crore ha** (about **46%** of reporting area, about **43%** of geographical area); **forests** about **22–23%** (recorded); the rest is barren, non-agricultural use, pastures, fallows, etc. **Gross cropped area (GCA)** counts land as many times as it is sown in a year: about **21.9 crore ha**. **Cropping intensity = GCA ÷ NSA × 100 ≈ 155%** (higher where irrigation allows two or three crops, e.g. Punjab ~190%). About **55%** of net sown area is **irrigated**; the rest is **rain-fed**.
- **India's cropping pattern: foodgrains** still take the largest share of GCA (about **60%**, down from about **75%** in 1950–51). **Rice** has the largest area (about **4.7 crore ha**), then **wheat** (about **3.1 crore ha**). The share of **commercial/cash crops** (cotton, sugarcane, oilseeds) and of **horticulture** has risen; **coarse cereals (millets)** lost area to rice and wheat after the Green Revolution. **Horticulture production (~36 crore t)** now **exceeds foodgrains** (since about **2012–13**).
- **What decides the cropping pattern? Geographical**: soil, rainfall, temperature, **irrigation** (canals and tube-wells made Punjab–Haryana rice–wheat). **Economic**: **prices and MSP**, assured **procurement** (mainly rice and wheat), input costs, market access, farm size. **Institutional and technological**: **HYV seeds**, credit, land tenure, research, government schemes. **Cropping systems**: **mono-cropping**, **multiple cropping** (two or more crops a year), **mixed cropping** (two crops together in one field, as insurance), **intercropping** (in rows), **relay cropping**, **crop rotation** (e.g. a legume after a cereal to restore nitrogen).

2. Production: the record

Crop group	2024–25 (final)	2025–26 (3rd advance estimate)

Total foodgrains	**357.7 MT** (record, +8%)	**376.6 MT** (new record)
Rice	150.2 MT	154.0 MT
Wheat	117.9 MT	120.7 MT
Maize	43.4 MT	55.1 MT (ethanol demand)
Pulses	25.7 MT	Gram 12.5, tur 3.6 MT
Oilseeds	43.0 MT	43.1 MT
Sugarcane / cotton	454.6 MT / 297 lakh bales	500.1 MT / 290 lakh bales

- **Long view:** foodgrains rose from about **51 MT (1950–51)** to about **376 MT (2025–26)**, roughly **seven times**, while area rose only modestly: the gain came mostly from **yield**. India is now a **net exporter** of food (rice exports **largest in the world**), yet still **imports edible oils** (about **55–60%** of need) and some **pulses**. India is **first** in **milk** (about **24%** of world output), **pulses** and **jute**; **second** in **rice, wheat, sugarcane, fruits and vegetables**, **cotton**.
- **Who estimates?** The **Directorate of Economics and Statistics (DES)**, Ministry of Agriculture, releases **advance estimates** and a **final estimate** each year (now **three** advance estimates). **Agricultural Census** (land holdings) every **five years**; the **Situation Assessment Survey** of farm households is done by the **NSO**.

3. Productivity: why low, how to raise

Why yields are low and what is being done

Cause of low yield	**Policy response**
Small, fragmented holdings (average about **1.08 ha**; **86%** small and marginal, Agri Census 2015–16)	**10,000 FPOs** (2020), cooperative farming, land-record digitisation (**SVAMITVA**, AgriStack), consolidation
Rain-fed farming on about **45%** of net sown area; monsoon risk	**PMKSY (2015)**: 'Har Khet Ko Pani', '**Per Drop More Crop**' (micro-irrigation); watershed development
Poor seeds, soil fatigue, imbalanced fertiliser (too much urea)	**Soil Health Card (2015)**, **neem-coated urea**, **nano urea**, new seed varieties (**109 varieties released 2024**), **PM-PRANAM**
Low mechanisation, credit and investment; post-harvest losses	**SMAM** (custom hiring), **KCC**, **Agri Infra Fund (₹1 lakh cr, 2020)**, cold chains, **e-NAM (2016)** for better prices
Crop-specific gaps (pulses, oilseeds, cotton)	**NMEO–Oil Palm (2021)**, **NMEO–Oilseeds (2024)**, **Mission for Aatmanirbharta in Pulses (2025)**, **PM Dhan-Dhaanya Krishi Yojana (2025)** for **100 low-yield districts**

- **Yields compared:** India's **foodgrain yield** rose from about **0.5 t/ha (1950–51)** to about **2.5–2.6 t/ha** now, but **rice and wheat yields** are still well **below China's** and the world's best; **pulses** yield is under **1 t/ha**. Big regional gaps: **Punjab** has India's highest wheat and rice yields; eastern States and rain-fed areas lag.

Green Revolution (1966–67) raised yields mainly in **wheat**, in **irrigated north-west**, causing **regional and crop imbalance**, groundwater depletion and soil damage.

- **Andhra Pradesh:** the **Godavari–Krishna deltas** are a **rice bowl** with yields above the national average; AP leads India in **aquaculture (shrimp)**, **oil palm** (area and output), **chillies** (Guntur) and is among the top in **mango, papaya, tomato (Madanapalle), banana, turmeric**; **Rayalaseema** grows **groundnut, millets, horticulture** under rain-fed conditions. Agriculture and allied sectors give about **a third of AP's GSVA**, higher than the national share, mainly because of **fisheries and livestock**. **Rythu Seva Kendras**, **e-Crop** booking, **APCNF** natural farming and **Annadata Sukhibhava** support farmers.

OTHER ANGLES ASKED IN EXAMS

Other angles asked: (1) **Cropping intensity formula** (GCA ÷ NSA × 100) and why irrigation raises it. (2) **Production = area × yield**: a question may give two of the three. (3) **Terms of trade** between agriculture and industry; **farm income**: NSO's survey (2018–19) put an agricultural household's average monthly income at about **₹10,218**. (4) **'Doubling Farmers' Income'** committee (**Ashok Dalwai**, 2016–18). (5) **Food security vs nutrition security**: the shift from cereals to **protein, fruit, vegetables, milk** changes the cropping pattern. (6) **Paddy in Punjab–Haryana** causes **groundwater depletion** and **stubble burning**; schemes pay farmers to **diversify** (crop diversification programme). (7) **International Year of Millets 2023**; '**Shree Anna**'. (8) Agriculture is a **State subject** (List II entry 14), though the Union runs most schemes.

EXAM TRAP

Cropping pattern = which crops, what share of area; production = total output; productivity = yield per ha; production = area × yield; **NSA ~14 crore ha (~46% reporting, ~43% geographical); GCA ~21.9 crore ha; cropping intensity ~155% (Punjab ~190%); ~55% NSA irrigated**; **foodgrains ~60% of GCA (75% in 1950–51); rice largest area (~4.7 crore ha), wheat ~3.1; horticulture (~36 crore t) > foodgrains since ~2012–13**; **factors: geographical, economic (MSP, procurement), institutional, technological; systems: mono, multiple, mixed, inter, relay, rotation**; **2024–25 final: 357.7 MT (rice 150.2, wheat 117.9, maize 43.4, pulses 25.7, oilseeds 43.0); 2025–26 3rd AE: 376.6 MT (rice 154.0, wheat 120.7, maize 55.1); 1950–51: ~51 MT**; **India 1st milk (~24%), pulses, jute; 2nd rice, wheat, sugarcane, F&V, cotton; largest rice exporter; edible oil imports ~55–60%**; **DES advance estimates; Agri Census every 5 years (holding ~1.08 ha, 86% small/marginal, 2015–16)**; **yield ~0.5 → ~2.5 t/ha; pulses < 1 t/ha; Punjab highest**; **PMKSY 2015, SHC 2015, e-NAM 2016, AIF 2020, 10,000 FPOs, NMEO-OP 2021, NMEO-Oilseeds 2024, Pulses mission 2025, PM Dhan-Dhaanya 2025 (100 districts)**; **NSO income ₹10,218/month (2018–19); Dalwai DFI**; **AP: delta rice, aquaculture, oil palm, chillies; agri ~1/3 of GSVA**.

Test yourself: 8 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:
1. Cropping intensity is calculated as net sown area divided by gross cropped area, multiplied by 100.
2. India's cropping intensity is about 155%.
3. Rice occupies the largest cropped area among all crops in India.

4. Horticulture production in India now exceeds foodgrain production.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: B. Statement 1 is reversed: cropping intensity = GCA ÷ NSA × 100. Statements 2, 3 and 4 are correct.

PRACTICE

Match List-I (term) with List-II (meaning):

List-I (term)	List-II (meaning)
A. Mixed cropping	1. Growing two or more crops on the same land in one year, one after another
B. Crop rotation	2. Growing two crops together in a field without fixed rows, as insurance
C. Multiple cropping	3. Growing crops in definite row patterns in the same field
D. Intercropping	4. Changing crops in a planned sequence over seasons to keep soil fertile

(1) A → 2 B → 1 C → 4 D → 3

(2) A → 2 B → 4 C → 1 D → 3

(3) A → 3 B → 4 C → 1 D → 2

(4) A → 1 B → 4 C → 2 D → 3

Answer: (2) Mixed: together, as insurance; rotation: planned sequence; multiple: several crops a year; intercropping: in rows.

PRACTICE

Which one of the following statements about India's foodgrain production is NOT correct?

A. The final estimate for 2024–25 put foodgrain output at about 357.7 million tonnes.

B. Foodgrain output has risen about seven times since 1950–51, mostly because of higher yields.

C. India is self-sufficient in edible oils and exports most of its oilseed output.

D. Production estimates are released by the Directorate of Economics and Statistics of the Agriculture Ministry.

Answer: C. India imports about 55–60% of its edible oil needs; hence the National Missions on Edible Oils.

PRACTICE

Assertion (A): After the Green Revolution, the area under coarse cereals declined in many parts of India.
Reason (R): High-yielding varieties, assured irrigation and MSP-backed procurement made rice and wheat more profitable.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Both true; R explains A: prices, procurement and HYVs pulled area from millets to rice and wheat.

PRACTICE

A district has a net sown area of 2 lakh ha. In a year 1.2 lakh ha of it is sown twice and the rest once. What is its cropping intensity?

A. 120%

B. 140%

C. 160%

D. 180%

Answer: C. GCA = 2 + 1.2 = 3.2 lakh ha; intensity = 3.2 ÷ 2 × 100 = 160%.

PRACTICE

Consider the following statements:

I. Agriculture employs about 45% of India's workers but contributes only about 16–18% of GVA.

II. The average size of an operational holding in India is more than 2 hectares.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Answer: A. Only I. The average holding is about 1.08 ha (Agri Census 2015–16), and 86% are small and marginal.

PRACTICE

Consider the following pairs (scheme – main aim):

1. PMKSY – 'Per Drop More Crop' micro-irrigation

2. NMEO–Oil Palm – raising edible-oil output

3. PM Dhan-Dhaanya Krishi Yojana – 100 low-productivity districts

4. e-NAM – crop insurance against natural calamities

How many of the above pairs are correctly matched?

A. Only one pair

B. Only two pairs

C. Only three pairs

D. All four pairs

Answer: C. Pair 4 is wrong: e-NAM is an online trading platform for farm produce; crop insurance is PMFBY.

PRACTICE

Arrange the following in chronological order:

1. e-NAM launched

2. PM Gram Sadak Yojana launched

3. NMEO–Oil Palm launched

4. Soil Health Card scheme launched

Select the correct order:

A. 2, 4, 1, 3

B. 4, 2, 1, 3

C. 2, 1, 4, 3

D. 2, 4, 3, 1

Answer: A. PMGSY 2000 → Soil Health Card 2015 → e-NAM 2016 → NMEO–Oil Palm 2021.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the three terms (cropping pattern, production, productivity) with one example each, land use numbers, cropping intensity formula and value. 2. Write the cropping-pattern shares, its causes and the cropping systems. 3. Write the production table (2024–25, 2025–26) and India's world ranks. 4. Write causes of low yield with one scheme each, and AP's farm profile.

TIMED PRACTICE · Economy: 30 questions in APPSC pattern (60 minutes)

UNDERSTAND IT FIRST

How to take this test: set a timer for **60 minutes**. Close all notes. **Write your choice on paper first**, then tap to check. Score: **+1** correct, **−1/3** wrong. **Target: 22 or more.** The questions cover the Economy of **Days 27–41**: national income, sectors and jobs, growth and development, planning and NITI, 1991 reforms, fiscal federalism, money, RBI, banking, inflation, fiscal policy, taxes and GST, the Budget, BOP and FDI, and agriculture.

PRACTICE

1. Consider the following statements:
1. GNP equals GDP plus net factor income from abroad.
2. National income is NNP at factor cost.
3. Factor cost equals market price minus indirect taxes plus subsidies.
4. Transfer payments such as old-age pensions are included in national income.
Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. Transfer payments are not payments for current production, so they are excluded. 1–3 are correct.

PRACTICE

2. If nominal GDP is ₹300 lakh crore and real GDP is ₹240 lakh crore, the GDP deflator is:

A. 80

B. 120

C. 125

D. 140

Answer: C. Deflator = nominal ÷ real × 100 = 300 ÷ 240 × 100 = 125.

PRACTICE

3. Match List-I (concept) with List-II (associated name):

List-I (concept)	List-II (associated name)
A. Big push	1. Ragnar Nurkse
B. Vicious circle of poverty	2. Albert Hirschman
C. Unbalanced growth	3. Rosenstein-Rodan
D. Stages of growth (take-off)	4. W.W. Rostow

(1) A → 3 B → 1 C → 2 D → 4

(2) A → 1 B → 3 C → 2 D → 4

(3) A → 3 B → 2 C → 1 D → 4

(4) A → 3 B → 1 C → 4 D → 2

Answer: (1) Big push: Rosenstein-Rodan; vicious circle: Nurkse; unbalanced: Hirschman; take-off: Rostow.

PRACTICE

4. Which one of the following is NOT a dimension of the UNDP's Human Development Index?

A. A long and healthy life

B. Access to knowledge

C. A decent standard of living (GNI per capita, PPP)

D. Infant mortality rate

Answer: D. HDI uses health, education and income. IMR is part of PQLI, not HDI.

PRACTICE

5. Consider the following statements:

I. Disguised unemployment means workers whose marginal productivity is zero, common in agriculture.

II. The unemployment rate is calculated as unemployed persons divided by the total population.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Answer: A. Only I. UR = unemployed ÷ labour force, not population.

PRACTICE

6. Arrange the following in chronological order:

1. Bombay Plan

2. Planning Commission set up

3. National Development Council set up

4. NITI Aayog set up

Select the correct order:

A. 1, 3, 2, 4

B. 2, 1, 3, 4

C. 1, 2, 3, 4

D. 1, 2, 4, 3

Answer: C. Bombay Plan 1944 → Planning Commission 1950 → NDC 1952 → NITI 2015.

PRACTICE

7. Which Five-Year Plan is correctly matched with its focus?

A. First Plan – heavy industry (Mahalanobis model)

B. Second Plan – agriculture (Harrod–Domar model)

C. Eleventh Plan – faster and more inclusive growth

D. Fifth Plan – sustainable growth

Answer: C. 1st: agriculture (Harrod–Domar); 2nd: heavy industry (Mahalanobis); 5th: garibi hatao; 11th: inclusive; 12th: sustainable.

PRACTICE

8. Assertion (A): The 1991 reforms included stabilisation measures that were short-term and demand-side.
Reason (R): Stabilisation was linked to the IMF, while structural adjustment was linked to the World Bank.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but R (which agency backed which part) does not explain why stabilisation was short-term and demand-side.

PRACTICE

9. Consider the following statements about the 1991 crisis and reforms:
1. Foreign exchange reserves had fallen to about two weeks of imports.
2. Gold was pledged with the Bank of England and the Bank of Japan.
3. The rupee was devalued in July 1991.
4. Full capital account convertibility was introduced in 1994.
Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. 1994 brought CURRENT account convertibility; capital account is still only partly convertible.

PRACTICE

10. Under Article 270 of the Constitution, which taxes form the divisible pool shared with States?

A. All taxes levied by the Union including cesses and surcharges

B. All Union taxes except cesses and surcharges (and a few specific ones)

C. Only income tax and corporation tax

D. Only customs duties

Answer: B. Since the 80th Amendment (2000), all Union taxes are shareable except cesses, surcharges (Art 271) and a few others.

PRACTICE

11. Consider the following pairs (Finance Commission – chairperson):
1. First – K.C. Neogy
2. Fourteenth – Y.V. Reddy
3. Fifteenth – N.K. Singh
4. Sixteenth – Vijay Kelkar
How many of the above pairs are correctly matched?

A. Only one pair

B. Only two pairs

C. Only three pairs

D. All four pairs

Answer: C. The 16th FC is chaired by Arvind Panagariya; Kelkar chaired the 13th.

PRACTICE

12. Which one of the following is the most liquid measure of money supply in India?

A. M1

B. M2

C. M3

D. M4

Answer: A. M1 (currency + demand deposits + other deposits with RBI) is the narrowest and most liquid; M4 the least.

PRACTICE

13. If the RBI raises the Cash Reserve Ratio, what happens to the money multiplier and broad money, other things remaining the same?

A. Both rise

B. Multiplier rises, broad money falls

C. Both fall

D. No effect on either

Answer: C. A higher CRR leaves banks less to lend, so the multiplier (M3 ÷ M0) and M3 both fall.

PRACTICE

14. Consider the following statements about the Monetary Policy Committee:
1. It has six members, three from the RBI and three nominated by the Government.
2. The Governor has a casting vote in case of a tie.
3. The inflation target is set by the RBI itself every year.
4. It must meet at least four times a year.
Which of the above are correct?

A. 1, 2 and 4 only

B. 1 and 3 only

C. 2 and 4 only

D. 1, 2, 3 and 4

Answer: A. The target (CPI 4% ± 2) is set by the Government, in consultation with the RBI, for five years.

PRACTICE

15. Match List-I (policy tool) with List-II (description):

List-I (policy tool)	List-II (description)
A. Standing Deposit Facility	1. Banks borrow overnight at repo + 0.25%
B. Marginal Standing Facility	2. Absorbs liquidity from the system
C. Statutory Liquidity Ratio	3. Floor of the policy corridor, absorbs funds without collateral
D. Open market sale of G-secs	4. Share of deposits banks keep in cash, gold or approved securities

(1) A → 1 B → 3 C → 4 D → 2

(2) A → 3 B → 1 C → 2 D → 4

(3) A → 3 B → 1 C → 4 D → 2

(4) A → 4 B → 1 C → 3 D → 2

Answer: (3) SDF: floor; MSF: repo + 0.25; SLR: cash/gold/G-secs; OMO sale: absorbs liquidity.

PRACTICE

16. Consider the following statements:

I. Fourteen banks were nationalised in July 1969 and six more in April 1980.

II. Regional Rural Banks are owned by the Centre, sponsor bank and State in the ratio 50 : 35 : 15.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Answer: C. Both are correct.

PRACTICE

17. Which one of the following statements about payments banks is correct?

A. They can give loans up to ₹2 lakh.

B. They can accept deposits up to ₹2 lakh per customer but cannot lend.

C. They must give 75% of loans to the priority sector.

D. They were recommended by the Narasimham Committee in 1991.

Answer: B. Payments banks (Nachiket Mor 2013; licences 2015) take deposits up to ₹2 lakh but cannot lend; the 75% PSL rule applies to small finance banks.

PRACTICE

18. A situation of high inflation along with stagnant output and high unemployment is called:

A. Disinflation

B. Skewflation

C. Reflation

D. Stagflation

Answer: D. Stagflation = inflation + stagnation + unemployment.

PRACTICE

19. Consider the following statements about price indices in India:

1. WPI is compiled by the Office of the Economic Adviser, DPIIT, and covers only goods.

2. CPI (Combined) is compiled by the NSO and food has the largest weight in it.

3. The RBI's inflation target is based on WPI.

4. CPI for Industrial Workers is used to revise dearness allowance.

Which of the above are correct?

A. 1, 2 and 4 only

B. 1 and 2 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. The RBI targets CPI (Combined) inflation, not WPI.

PRACTICE

20. Inflation is generally beneficial to which one of the following?

A. Pensioners on a fixed pension

B. Lenders who gave fixed-rate loans

C. Borrowers who repay fixed sums

D. Savers holding cash

Answer: C. Borrowers repay in money that is worth less; fixed-income earners, lenders and cash savers lose.

PRACTICE

21. If the fiscal deficit is ₹16 lakh crore and interest payments are ₹13 lakh crore, the primary deficit is:

A. ₹29 lakh crore

B. ₹13 lakh crore

C. ₹3 lakh crore

D. ₹16 lakh crore

Answer: C. Primary deficit = fiscal deficit − interest = 16 − 13 = ₹3 lakh crore.

PRACTICE

22. Consider the following statements about fiscal rules:
I. The FRBM Act, 2003 set the Centre's fiscal deficit target at 3% of GDP.
II. The N.K. Singh Committee (2017) recommended general government debt of 60% of GDP (40% Centre, 20% States).
Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Answer: C. Both are correct.

PRACTICE

23. Which one of the following taxes is NOT subsumed in GST and remains outside it?

A. Central excise on most manufactured goods

B. Service tax

C. State VAT on petrol and diesel

D. Entertainment tax levied by States

Answer: C. Petrol, diesel, ATF, crude, natural gas and alcohol for drinking remain outside GST; States levy VAT on them.

PRACTICE

24. Consider the following statements about the GST Council:

1. It is chaired by the Union Finance Minister.

2. The Centre has one-third and the States two-thirds of the weight of votes.

3. A decision needs three-fourths of the weighted votes.

4. The Supreme Court held in 2022 that its recommendations are binding on Parliament.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. Mohit Minerals (2022): recommendations are persuasive, not binding. 1–3 are correct.

PRACTICE

25. Arrange the following in chronological order:

1. GST introduced

2. Wealth tax abolished

3. Income-tax Act 2025 comes into force

4. GST 2.0 with 5% and 18% main slabs

Select the correct order:

A. 2, 1, 3, 4

B. 1, 2, 4, 3

C. 2, 1, 4, 3

D. 2, 4, 1, 3

Answer: C. Wealth tax abolished 2015 → GST 2017 → GST 2.0 Sept 2025 → new IT Act April 2026.

PRACTICE

26. Consider the following pairs (Article – budget provision):

1. Article 112 – Annual Financial Statement

2. Article 113 – Demands for grants

3. Article 116 – Vote on account

4. Article 267 – Consolidated Fund of India

How many of the above pairs are correctly matched?

A. Only one pair

B. Only two pairs

C. Only three pairs

D. All four pairs

Answer: C. Art 267 is the Contingency Fund; the Consolidated Fund is Art 266.

PRACTICE

27. In the Union Budget 2026–27 (Budget Estimates), the fiscal deficit target is about:

A. 3.0% of GDP

B. 4.3% of GDP

C. 4.8% of GDP

D. 5.6% of GDP

Answer: B. FD 4.3% of GDP (₹16.96 lakh crore); revenue deficit 1.5%; debt about 55.6% of GDP.

PRACTICE

28. Assertion (A): India usually runs only a small current account deficit despite a large merchandise trade deficit.

Reason (R): Large surpluses from services exports and remittances offset much of the goods deficit.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Both true; R explains A: invisibles surplus offsets the goods deficit (CAD ~0.6% of GDP in 2024–25 and 2025–26).

PRACTICE

29. Which one of the following statements about FDI and FPI is NOT correct?

A. An investment of 10% or more in a listed company is treated as FDI.

B. FPI is often called 'hot money' because it can leave quickly.

C. Press Note 3 (2020) requires government approval for FDI from countries sharing a land border with India.

D. FDI is fully banned in the defence sector.

Answer: D. Defence allows FDI up to 74% automatic and above that through the government route.

PRACTICE

30. Consider the following statements about Indian agriculture:
1. Foodgrain production in 2024–25 (final estimate) was about 357.7 million tonnes.
2. India is the world's largest exporter of rice.
3. India meets most of its edible-oil needs from domestic production.
4. Average farm holding size is about 1.08 hectares (Agricultural Census 2015–16).
Which of the above are correct?

A. 1, 2 and 4 only

B. 1 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. India imports about 55–60% of its edible oil. The rest are correct.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

After the test: score = correct − (wrong ÷ 3). For each wrong answer write the question number, the correct fact in one line and its topic (national income, sectors, growth, planning, reforms, federal finance, money, RBI, banking, inflation, fiscal, tax/GST, Budget, BOP/FDI, agriculture). Re-read only those parts.

NOTEBOOK PAGE · Day 41: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Transport and communication

Area	Write
Roads	Network rank and length; NH share and growth (2014 → 2025); NHAI; Golden Quadrilateral; NS–EW corridors and Jhansi; Bharatmala; PMGSY; NH numbering rule; NH-44, NH-16; BRO
Railways	1853 first train; 1924 separate budget, 2017 merger; route km, rank, electrification; gauges; UNESCO hill railways; 18 zones incl. SCoR (Vizag, 1 June 2026); DFCs; Vande Bharat, bullet train, Kavach, Chenab bridge; factories; KK line
Water, air, pipelines	NW Act 2016; NW-1 to NW-5; IWAI; cargo growth; 12 major ports (west and east) with one feature each; Vadhavan; Sagarmala; AAI, DGCA, UDAN, Air India; pipelines (Naharkatiya, HBJ, Urja Ganga, cross-border)
Communication and AP	Post 1854, PIN 1972, telegraph end 2013, 5G 2022, Telecom Act 2023, TRAI; AIR, Akashvani, Doordarshan, Prasar Bharati; AP's NH-16, rail junctions, SCoR, ports, airports incl. Bhogapuram, NW-4

ECONOMY: Indian agriculture

- Define cropping pattern, production, productivity; production = area × yield; land use numbers (NSA, GCA, irrigation); cropping intensity formula and ~155%; foodgrain share of GCA then and now; rice and wheat areas; horticulture vs foodgrains; factors behind cropping pattern; cropping systems; production table 2024–25 and 2025–26; India's world ranks; edible-oil imports; DES estimates; yield 1950 vs now; causes of low yield with one scheme each; Green Revolution effects; AP profile.

PRACTICE: Economy test

Write your score and the topic of each wrong answer (national income, growth, planning, reforms, federal finance, money and RBI, banking, inflation, fiscal, taxes, Budget, BOP, agriculture); re-read those parts.